

**MATH 2101 LINEAR ALGEBRA I-PRACTICE PROBLEMS (UPDATED
1 NOV)**

1. VECTOR SPACES

- (1) Let $v_1, \dots, v_n \in \mathbb{R}^n$ be column vectors. Show that $\{v_1, \dots, v_n\}$ is linearly independent if and only if the $n \times n$ matrix $(v_1 \dots v_n)$ is invertible.
- (2) (a) Compute the determinant of the matrix

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 3 \\ 2 & 1 & 1 & 0 \\ 0 & -1 & 2 & 1 \end{pmatrix}.$$

- (b) Use Questions 1 and 2(a) to determine if the vectors $(1 \ 1 \ 2 \ 0)^T, (0 \ 2 \ 1 \ -1)^T, (1 \ 0 \ 1 \ 2)^T, (0 \ 3 \ 0 \ 1)^T$ are linearly independent.
- (3) Describe all possible row vectors v in $\mathbb{R}^3$ such that

$$\{(1 \ 0 \ 0), (0 \ 1 \ 0), v\}$$

forms a basis.

- (4) Find all possible values of a so that

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} a \\ 3 \\ 1 \end{pmatrix} \right\}$$

forms a basis for $\mathbb{R}^3$.

- (5) Show that $\{v_1, v_2\}$ is a basis for a vector space V if and only if $\{v_1 + v_2, v_1 - v_2\}$ is a basis for a vector space V . (Hint: you only need to show $\{v_1, v_2\}$ is linearly independent if and only if $\{v_1 + v_2, v_1 - v_2\}$ is linearly independent.)
- (6) Let $v_1, \dots, v_r$ be a set of linearly independent vectors in a vector space V . Suppose $v' \notin \text{span}(\{v_1, \dots, v_r\})$. Then $\{v_1, \dots, v_r, v'\}$ is a set of linearly independent vectors.
- (7) Use the previous question to find a basis for $\mathbb{R}^3$ that contains the vector $(1 \ 2 \ -1)^T$.

2. LINEAR TRANSFORMATIONS

- (1) Determine if the following functions are linear transformations. Explain your answer.
- (a) Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$; $T(x_1, x_2, x_3) = (x_1, x_2)$.
- (b) Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$; $T(x_1, x_2) = (x_1, x_2 x_3, x_3)$.
- (c) Let M_2 be the vector space of 2×2 matrices. Define $T : M_2 \rightarrow M_2$; $T(A) = A^T$.

- (d) Define $T : M_2 \rightarrow M_2$; $T(A) = A + I_2$.
- (e) Let V be the vector space of functions from $\mathbb{R}$ to $\mathbb{R}$. Define $T : V \rightarrow \mathbb{R}$ by $T(f) = f(1)$. (Here $f(1)$ means to evaluate f at 1.)
- (2) Compute $R(T)$ and $N(T)$ of linear transformations T in Problem 1. (No need to compute if T is not a linear transformation.)
- (3) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation. It is known that $T((1, 0, 1)) = (0, 1)$ and $T((0, 1, 0)) = (0, 1)$. Describe all the possibilities of T in terms of a basis of $\mathbb{R}^3$.
- (4) Let $T : V \rightarrow W$ and $U : W \rightarrow X$ be linear transformations. Prove that $\text{rank}(U \circ T) \leq \text{rank}(T)$ and $\text{nullity}(U \circ T) \geq \text{nullity}(T)$.
- (5) Prove or disprove the following statements:
 - (a) Let $T : V \rightarrow W$ be a linear transformation and let $U : W \rightarrow X$ be an invertible linear transformation. Then $\text{rank}(U \circ T) = \text{rank}(T)$ and $\text{nullity}(U \circ T) = \text{nullity}(T)$.
 - (b) Let $T : V \rightarrow W$ and $U : W \rightarrow X$ be linear transformations. Then $\text{rank}(U \circ T) \leq \text{rank}(U)$.
 - (c) Let $T : V \rightarrow W$ and $U : W \rightarrow X$ be linear transformations. Then $\text{nullity}(U \circ T) \geq \text{nullity}(U)$.
- (6) Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$; $T(x_1, x_2, x_3, x_4) = (x_1 + x_2 + x_3 + x_4, x_2 + x_3, x_1 + x_4, x_1 - x_2 - x_3 + x_4)$. Find $\text{rank}(T)$ and $\text{nullity}(T)$. Is T injective?
- (7) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a linear transformation. Show that $T' : \mathbb{R}^6 \rightarrow \mathbb{R}^2$ given by

$$T'((x_1, x_2, x_3, x_4, x_5, x_6)) = (T((x_1, x_2, x_3)), T((x_4, x_5, x_6)))$$
 is a linear transformation.
- (8) Let V be the set of differentiable functions from $\mathbb{R}$ to $\mathbb{R}$. Define $T : V \rightarrow B$ given by $T(f) = f'$. Is T injective?

3. MATRIX REPRESENTATIONS

- (1) Let $P_2(\mathbb{R})$ be the vector space of all polynomials of degree less than or equal to 2 (with coefficients in $\mathbb{R}$). Let $\beta = \{1, x, x^2\}$ and let $\beta' = \{1 + x, x + x^2, 1 + x^2\}$. Let $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be the map of differentiation i.e. $T(f) = \frac{df}{dx}$. Find the matrix representation $[T]_{\beta}^{\beta'}$. Is T invertible?
- (2) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by

$$T(x_1, x_2, x_3) = (x_1 + x_2 - x_3, x_2 + x_3, -x_1 - x_2 - x_3).$$

Is T invertible? If so, find T^{-1} .

- (3) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by:

$$T(x_1, x_2, x_3) = (x_1 + 3x_2 - x_3, x_1 + 2x_3, x_2 + 4x_3)$$

and let $S : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ given by:

$$S(x_1, x_2, x_3) = (x_2 + x_3, x_3 + x_1, x_1 + x_2, x_1 + x_2 + x_3).$$

Find $(S \circ T)(x_1, x_2, x_3)$.